Usage of mapping properties on the semantic web to align the existing variety in the semantics of Lyme disease

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Abstract

Disease entities are connected across biomedical semantic resources through a variety of mapping relations, whose semantics and usage differ between data providers. As a result, a single disease is not described in one place in the biomedical semantic web but across multiple resources, each providing partial and complementary representations. These differences influence how disease knowledge is interpreted when representations from multiple sources are combined.

In this paper, we analyse how mapping choices shape disease-level integration by examining the representation of Lyme disease across RDF-based biomedical resources. Starting from the Orphanet Rare Disease Ontology (ORDO), we iteratively follow inter-resource links to identify disease entities and mappings across 17 connected resources, eight of which are openly available for quantitative analysis.

Our results show that inter-resource links rely predominantly on cross-reference properties such as oboInOwl:hasDbXref, while SKOS mapping relations are used more selectively and unevenly across resources.

At the disease level, entities defined at different clinical or modelling scopes are at times linked by equivalence-oriented relations, shaping how distinctions preserved within individual resources are carried into integrated representations.

To support the interpretation of these complex mapping structures and workflow, we develop a first visualization prototype that provides an integrated view of disease entities, mapping properties, and resource coverage. Overall, the study illustrates how mapping practices influence integrated disease representations and underlines the importance of making mapping semantics explicit for FAIR reuse of biomedical knowledge.

Keywords

Lyme disease, Mapping properties, Rare diseases, Metadata, Interoperability

1. Introduction

1.1. Rare disease data integration and FAIR challenges

Data on rare diseases are inherently fragmented, heterogeneous, and distributed across multiple biomedical resources [1, 2]. Information about a single disease is typically spread over clinical terminologies, research ontologies, classification systems, and domain-specific vocabularies, each capturing various aspects of the disease. As a result, integrating, comparing, and reusing rare disease data increasingly relies on biomedical ontologies and their interconnections [3].

In this context, the FAIR principles have become a central reference for structuring and sharing rare disease data. Ontologies play a key role in making disease data findable and interoperable, but this role critically depends on the quality and semantics of the mappings that connect disease entities

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across resources. Without explicit and interpretable mappings, the integration of heterogeneous disease representations remains fragile, even when data are formally FAIR-compliant [4].

1.2. Mappings as an underestimated source of semantic bias

Mappings between biomedical ontologies are often treated as technical artifacts intended to enable identifier alignment or cross-resource navigation [3, 5]. In practice, however, mapping relations represent a complex domain and set of knowledge.

The choice of mapping properties, such as `oboInOwl:hasDbXref`, OWL properties, SKOS relations, or ontology-specific properties, directly influences how medical entities are interpreted when data from multiple resources are combined [6]. Despite this, mappings are rarely analysed as semantic objects in their own right, and their interpretability is often taken for granted. As a consequence, conceptual differences between disease representations may be obscured or flattened during data integration, introducing semantic bias that remains largely implicit.

1.3. Lyme disease as a case study

Lyme disease provides a particularly suitable case study for analysing semantic mappings in the context of the Semantic Web. Lyme disease is caused by bacteria belonging to the *Borrelia burgdorferi sensu lato* complex, and is the most frequently occurring tick-borne infection in the Northern Hemisphere [7]. The incidence of LD varies widely in Europe [8] In the Netherlands, approximately 27,000 new diagnoses are reported annually[9]. Most patients (around 25,500 per year) present with early localized infection, typically erythema migrans, while a smaller proportion (900 to 1,500 patients) develop disseminated LD [9, 10] There are multiple manifestations of disseminated LD affecting different organ systems. LD should not be considered a rare disease in the Netherlands, however it can have rare manifestations, such as Lyme carditis, which accounts for only 0.3-4% of LD cases

Although early manifestations generally respond well to antibiotic treatment, a variable proportion of patients (0 to 48% depending on definitions and follow-up duration) develop post-treatment Lyme disease syndrome (PTLDS)[11]. The distinction between early or disseminated Lyme disease and PTLDS, or long-lasting symptoms with, after, or attributed to LD, is clinically and conceptually important, as it may directly influence patient management.

In addition, there are multiple bacterial genospecies within the *Borrelia burgdorferi sensu lato* complex, which includes *Borrelia burgdorferi sensu stricto* as one of its members but should not be conflated with it. Variation among B. burgdorferi sensu .lato. strains plays a major role in the marked regional differences seen in the clinical manifestations of LD. [12]

These biological and clinical variations should be accurately and comprehensibly represented in biomedical resources, however, they are often unevenly reflected, with differences in how disease boundaries, manifestations, and subtypes are defined .

As a result, Lyme disease is represented through multiple, partially overlapping disease entities across ontologies and terminologies. This makes it a good example of a semantically interesting disease, where mapping choices play a decisive role in shaping how disease knowledge is connected and interpreted.

1.4. Contribution of this paper

In this paper, we present a systematic analysis of how Lyme disease entities are connected across biomedical ontologies through mapping relations. Using a cascade-based approach starting from ORDO [13], we identify and analyse mappings across multiple resources, focusing on the properties used to express these links and their distribution.

- Disease-level mapping analysis
- Quantitative overview of mapping properties
- Visualization prototype

Together, these contributions aim to highlight the importance of mapping choices in rare and complex disease data integration and to provide tools for researchers and practitioners seeking to understand disease representations across the biomedical semantic web.

2. Background

This section introduces the minimum semantic web concepts required to understand the analyses and results presented in this paper. It does not aim to provide a comprehensive overview of biomedical ontologies or standards, but rather to clarify how biomedical knowledge is represented, accessed, and connected across heterogeneous resources, and why mapping choices play a central role in disease integration.

2.1. Semantic Web resources for biomedical knowledge

Biomedical knowledge about diseases is not stored in a single, unified system. Instead, it is distributed across many semantic resources, each created to serve a specific purpose. Some resources focus on clinical documentation, others on epidemiology, research annotation, literature indexing, or computational modelling. As a consequence, the same disease may be described using different identifiers, names, scopes, or levels of detail depending on the context [3].

To support interoperability, many of these resources adopt semantic web principles, where knowledge is represented as explicitly linked data [14, 15, 16]. In this setting, diseases, genes, phenotypes, or classifications are treated as identifiable entities that can be connected across resources. However, not all semantic resources are ontologies in the strict sense. In practice, the biomedical semantic web includes a mix of formatrepresentations like ontologies, terminologies, classification systems, and thesaurusi, which differ in structure and semantic expressiveness.

This diversity is particularly pronounced in biomedicine because resources are developed worldwide by different communities : clinical organisations, research institutes, public health agencies, biocurators, developers, each addressing distinct needs. Rather than replacing one another, these resources coexist and must be related. Mappings provide the links that allow users and machines to navigate between them, making cross-resource discovery and integration possible.

2.2. RDF and SPARQL as a common representation layer

Disease entities and their mappings are exposed across biomedical ontologies using RDF-based representations, which allow heterogeneous resources to be accessed as a unified graph of identifiers and relations. RDF provides a minimal but sufficient data model to represent disease concepts, external references, and mapping assertions independently of internal ontology design. SPARQL enables the systematic extraction of disease entities and mapping relations by querying graph patterns rather than resource-specific schemas, making it possible to trace disease-level connections across multiple ontologies using shared identifiers and mapping properties. This combination constitutes the technical substrate underlying the cascade-based mapping analysis performed in this study. [17, 18]

2.3. Mapping properties: semantics, usage, and practical differences

Mappings can be expressed using specific mapping properties, which define how two entities from different resources are related. Crucially, these properties do not all mean the same thing. Some indicate strong semantic equivalence, while others merely signal a loose association or an external reference. Choose to use properties that come from shared vocabulatries such as SKOS or OWL improve interoperability. [19, 20]

Practitioners often face the challenge of selecting appropriate properties for integration: a simple cross-reference may not suffice when concepts have different scopes or levels of granularity, while equivalence assertions may be inappropriate when only approximate similarity exists. Inconsistent use

Table 1
Commonly used mapping properties and their typical interpretation

Mapping property	Intended meaning	Typical interpretation	Practical implication
`oboInOwl:hasDbXref`	Cross-reference to an external identifier	Indicates that two resources refer to related identifiers	No semantic equivalence implied, use with caution
`skos:exactMatch`	Concepts can be used interchangeably across systems	Strong equivalence	Suitable when concepts are semantically aligned
`skos:closeMatch`	Concepts are highly similar but not necessarily identical	Approximate equivalence	Useful for broader semantic overlap
`skos:broadMatch`	Target concept is broader than source	Hierarchical relation (broader)	Indicates generalisation
`skos:narrowMatch`	Target concept is narrower than source	Hierarchical relation (narrower)	Indicates specialisation
`skos:relatedMatch`	Concepts are related without hierarchical implication	Associative link	Low semantic commitment
`crossSpeciesExactMatch`	Equivalence across species-specific views	Strong equivalence in cross-species context	Useful for multi-organism datasets, context dependent

of these properties across resources can lead to ambiguities in integration outcomes, especially when downstream processing assumes particular semantics. [5]

In theory, stronger OWL-based relations such as owl:equivalentClass, owl:sameAs, or explicit hierarchical axioms could be used to represent disease mappings [21]. In practice, these properties are rarely applied to diseases, as they impose strict logical commitments that are difficult to justify in complex and heterogeneous biomedical contexts [19]. As a result, resources tend to rely on lightweight mapping properties or informal annotations, which are easier to assert but provide weaker semantic guarantees.

These mapping properties underlie much of the observed variability in disease integration and directly influence how semantic web resources can be combined, queried, and visualised. A clear understanding of their semantics and practical differences is therefore foundational for interpreting the integrated results presented later in this work.

3. Material & Methods

Using Lyme disease as a case study, we examined entities and mappings in 17 resources including MONDO , ORDO, DOID, ICD-9/10/11, MESH, NCIT, and UMLS [14, 13, 16, 22**?**]. SPARQL queries were employed to extract mapping relations, retrieve identifiers, and assess property usage such as oboinowl:hasDbXref, skos:exactMatch, skos:closeMatch. Our analysis quantified the proportion of mappings using each property and highlighted cases where mappings link distinct clinical subtypes, different manifestations, and technical difficulties in extracting all this information. In order to facilitate the interpretation of these complex relationships, we are developing a visualisation prototype that aims to present entities, mappings and metadata in a user-friendly interface, allowing researchers to easily explore disease coverage through linked data, diversity and alignment types.

3.1. Case study rationale and scope

Lyme disease is used in this study as a concrete disease instance to examine how a single condition is represented, identified, and linked across multiple biomedical ontologies. The disease is treated exclusively as an object of semantic analysis, without reference to its clinical history or epidemiology. All analyses are performed at the level of disease entities and their associated identifiers, definitions, and mappings.

In this study, we decide to start with the instantiation of Lyme disease represented in ORDO, which serves as the initial point of entry into the ontology network. From this starting entity, the study focuses on how equivalent or related disease entities are retrieved, aligned, and described across heterogeneous resources. The case study is used to trace how disease information propagates through mappings, rather than to assess the biomedical validity of the representations.

Practical constraints related to resource access, data formats, and licensing are considered as part of the methodological scope, as they directly affect which resources can be queried, integrated, or compared.

3.2. Resource selection and access

As a starting point, we selected the Orphanet Rare Disease Ontology (ORDO), which is widely considered a reference resource for rare disease representation [13]. ORDO provides structured identifiers and explicit mappings to other biomedical resources in an accessible OWL format, which facilitates querying and integration in semantic workflows. Additional resources were identified through mappings exposed by ORDO and by subsequently retrieved ontologies, following a cascade-based exploration strategy.

Resource availability was assessed using official ontological distributions and public registries, including BioPortal and BioRegistry [23, 24]. Following cascade analysis, resources were selected based on three criteria: availability of a distribution in RDF format (in English), license conditions allowing local analysis, and containing an entity with a direct or indirect mapping link from ordo.

Resources that required authentication, restricted licenses, or lacked an appropriate format downloadable, were excluded from direct querying but retained as referenced resources when they appeared as mapping targets. This distinction ensures consistency in data handling while preserving the ability to observe cross-resource links.

3.3. Mapping extraction strategy

Mappings were extracted using SPARQL queries applied to each selected ontology. The extraction focused on identifying explicit inter-ontology links expressed through dedicated mapping properties, including oboInOwl:hasDbXref and SKOS mapping relations.

Different retrieval strategies were applied depending on the role of the resource. For ORDO, disease entities were identified using label-based queries to locate the Lyme disease concept. For other resources, entity retrieval relied primarily on identifier-based queries, using disease codes referenced in mappings. Mapping-based retrieval was then used to federate across resources, enabling the identification of indirectly connected disease entities.

This approach allows the progressive expansion of the disease network without assuming direct equivalence between entities at retrieval time.

3.4. Graph construction and normalization

All retrieved files were imported into a GraphDB instance to build an RDF knowledge graph and enable integrated querying via a local SPARQL endpoint. This graph aggregated the accessible representations of Lyme disease and their interconnections across multiple ontologies. In the resources, some mappings linked directly to external URIs whereas others pointed to literal strings. To address this, a manual harmonization step was performed to align these literals with their corresponding identifiers when possible. This included distinguishing resolvable URIs from string-based identifiers, preserving original CURIE prefixes, and recording multiple identifier formats when they co-occurred. The resulting graph provides a structured representation of disease entities and mappings while maintaining the heterogeneity of the source data.

3.5. Representation of mappings using SSSOM

For the analysis and inspection, existing mapping properties were converted into a SSSOM tabular representation. This conversion was performed as a representation step with a SPARQL query, without introducing new mappings or modifying existing assertions. The objective of this conversion was to improve readability and facilitate comparison across resources by standardizing the representation of mapping source, target, and property. The SSSOM format is used here as a descriptive framework, not as a methodological proposal or validation mechanism.

Table 2
External mapping properties used by biomedical resources

Source	oboInOwl:hasDbXref	skos:exactMatch	skos:closeMatch	skos:broadMatch	skos:narrowMatch	skos:relatedMatch	crossSpeciesExactMatch
EFO	⊠	⊠	⊠	⊠	⊠	⊠	⊠
MONDO	⊠	⊠	⊠	⊠	⊠	⊠	⊠
DOID	⊠	⊠	⊠	⊠	⊠	⊠	–
ORDO	⊠	⊠	⊠	–	–	–	–
NCIT	⊠	–	–	–	–	–	–
ICD-10-CM	–	–	–	–	–	–	–
ICD-9-CM	–	–	–	–	–	–	–
MeSH	–	–	–	–	–	–	–

3.6. Visualization prototype

A visualization prototype was developed to make the structural complexity of disease representations and mappings explicit. The objective of the prototype is to provide an overview of how a single disease entity is connected to multiple resources and how different mapping properties are used across the network. The visualization is constructed from the integrated RDF graph and displays disease entities as nodes and mapping relations as edges, with visual differentiation based on mapping property types and source resources. The prototype is intended to support exploration and inspection of the mapping landscape. The visualization is not designed as a clinical decision tool, nor as a validated ontology alignment system. It serves solely as an exploratory aid to render the observed semantic structures visible.

4. Results

4.1. Retrieved resources and coverage

Using the cascade-based analysis strategy starting from ORDO, we identified a total of 17 distinct biomedical resources related to Lyme disease, including resources connected indirectly through secondary mappings. These resources represent a heterogeneous ecosystem of disease ontologies, terminologies, and reference vocabularies.

Among these, eight resources were openly accessible and downloadable in OWL or RDF/Turtle format, allowing direct querying and structural analysis. These resources are listed in Table 1 (available in the GitHub repository) and include: ORDO (Orphanet Rare Disease Ontology), MONDO (Mondo Disease Ontology), EFO (Experimental Factor Ontology), MeSH (Medical Subject Headings), DOID (Human Disease Ontology), ICD-9-CM, ICD-10-CM, and NCIT (National Cancer Institute Thesaurus).

In addition, nine related resources were identified through mappings but could not be directly downloaded as RDF/OWL under our access and licensing criteria. These are summarized in Table 2 (GitHub) and include SNOMED CT (international and US editions), MedDRA, ICD-10, ICD-11, UMLS, MedGen, NCIM, and GARD [25, 26, 27, 15, 28, 22, 29]. Although these resources play a significant role in disease representation and clinical coding, their restricted access or non-open licensing prevented their inclusion in the quantitative analyses.

Overall, the cascade method enabled the retrieval of multiple disease-related entities across heterogeneous resources, revealing a broad but uneven coverage of Lyme disease representations within the semantic web. Detailed lists of retrieved entities and their corresponding sources are provided in the GitHub repository.

4.2. Mapping property usage across ontologies

We analysed the usage of mapping properties across the eight openly accessible resources by counting all mapping assertions and their associated properties . This analysis was performed using a SPARQL query (GitHub: query-table-3.ttl) that qive the result : Absolute counts and relative proportions were computed per resource (Table with full results in GitHub).

The results, summarized in Table 2, show that mapping practices vary substantially between resources. Some ontologies rely on a single mapping property, while others employ a broader set of SKOS relations

to express inter-ontology links.

NCIT and ORDO primarily use oboInOwl:hasDbXref to represent external references. In NCIT, 100% of observed mappings rely on this property, while ORDO uses hasDbXref for 98.75% of mappings, with only a marginal presence of SKOS relations. This reflects a uniform, minimally expressive mapping strategy..

In contrast, MONDO, EFO, and DOID exhibit a richer and more diverse use of mapping properties. These resources combine hasDbXref with multiple SKOS relations, including skos:exactMatch, skos:closeMatch, skos:broadMatch, and skos:narrowMatch. For example, in MONDO, 47.36% of mappings are hasDbXref, 46.33% exactMatch, and 5.48% closeMatch, illustrating a broader inter-ontology linking strategy. DOID relies predominantly on hasDbXref (92.53%) but also includes smaller proportions of exactMatch, closeMatch, broadMatch, and narrowMatch. EFO shows the widest variety of mapping properties, including crossSpeciesExactMatch, reflecting the richest inter-ontology linking model among the resources analyzed.

By contrast, ICD-9-CM, ICD-10-CM, and MeSH do not expose external mappings in their official OWL distributions. As a result, no mapping properties were detected for these resources in this first global analysis.

Detailed counts and proportions of mapping properties per resource are provided in the GitHub repository in an SSSOM format.

4.3. Lyme disease entities and mappings

Focusing specifically on Lyme disease, we extracted all entities corresponding to Lyme disease across the identified resources, together with their associated mappings (GitHub: query-table-4.ttl). Each entity was identified using its native identifier, and where available, its label, definition, and external links.

Table 3
Extract of mapping results for Lyme disease across major biomedical terminologies and ontologies

Resource	Code	URI / URL	Label
DOID	DOID:11729	https://disease-ontology.org/?id=DOID:11729	Lyme disease
MONDO	MONDO:0019632	https://monarchinitiative.org/MONDO:0019632	Lyme disease
ORDO	ORPHA:91546	https://www.orpha.net/en/disease/detail/91546	Lyme disease
NCIM	C0024198	https://evsexplore.semantics.cancer.gov/evsexplore/concept/ncim/C0024198	Borrelia burgdorferi infection
ICD-11	1C1G	http://id.who.int/icd/entity/1600014919	Lyme borreliosis
EFO	EFO:0008510	http://www.ebi.ac.uk/efo/EFO_0008510	Lyme disease

Examples of retrieved identifiers include Orphanet_915646 (“Lyme disease”), MONDO:0019632 (“Lyme disease”), and DOID:11729 (“Lyme disease”), SCTID:48982009 (“Bannwarth syndrome”). The total number of distinct Lyme disease entities identified across all resources is reported in the GitHub tables, together with a representative excerpt included in the manuscript.

The analysis revealed substantial heterogeneity in how external identifiers are represented. For MeSH, ORDO references MeSH concepts using CURIE-like string literals (e.g. MeSH:D008193) without resolvable URIs, whereas MONDO uses both literal cross-references and resolvable URIs depending on the mapping property. For example, MONDO represents MeSH identifiers as string literals in hasDbXref, while using resolvable identifiers.org URIs in skos:exactMatch.

A similar pattern was observed for NCIT identifiers. MONDO exposes NCIT identifiers both as CURIE literals (e.g. NCIT:C45161) and as resolvable OBO Library URIs, whereas EFO includes multiple hasDbXref literals referring to the same NCIT concept, differing only in capitalization.

For ICD classifications, additional variability was observed. ORDO represents ICD-10 and ICD-11 codes as string literals (e.g. ICD-10:A69.2, ICD-11:1C1G), while MONDO uses a mixture of ICD-9, ICD-10-CM, and ICD-11 identifiers with different prefix conventions and, in some cases, resolvable WHO or BioPortal URIs. DOID and EFO similarly rely on string-based cross-references, sometimes duplicating identifiers across ICD versions.

Beyond identifiers, definitions associated with Lyme disease entities also vary across resources. For instance, EFO defines Lyme disease as "an infectious disease caused by Borrelia burgdorferi", while MeSH explicitly distinguishes between Lyme disease and Lyme neuroborreliosis, the latter being described as a nervous system infection caused by tick-borne spirochetes of the Borrelia burgdorferi group. NCIM provides a definition emphasizing geographical variation in causative species. These definitional differences are preserved in the extracted tables and illustrate the diversity of semantic scopes associated with Lyme disease entities.

4.4. Observed modelling divergences across resources

The extracted mappings further revealed contrasting modelling choices for clinically related entities associated with Lyme disease. One illustrative case concerns **Bannwarth syndrome**.

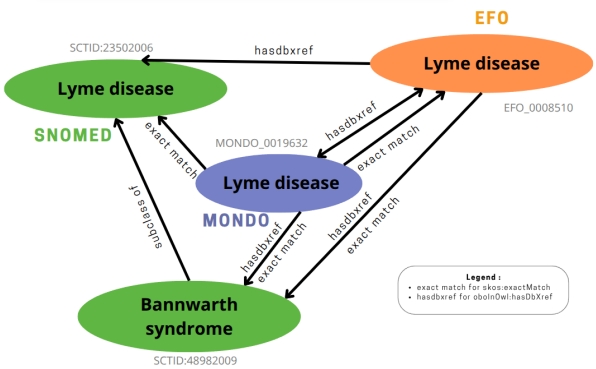

Figure 1: Visual representation of mapping properties between SNOMED CT, EFO and MONDO around Lyme disease

In SNOMED CT, Bannwarth syndrome (SCTID:48982009) is modelled as a clinical manifestation or subtype of Lyme disease (SCTID:23502006), establishing an explicit parent–child relationship. This structure reflects a hierarchical clinical differentiation between the general disease and a specific neurological presentation.

In contrast, EFO and MONDO align Bannwarth syndrome and Lyme disease using skos:exactMatch, asserting equivalence between the two entities. MONDO further links both concepts independently via exact matches, thereby flattening the hierarchical distinction present in SNOMED CT.

A comparable pattern was observed for MeSH entities. MeSH distinguishes between D008193 ("Lyme Disease") and D020852 ("Lyme Neuroborreliosis"), with the latter positioned as a child concept in the MeSH hierarchy and explicitly referring to nervous system involvement and a broader Borrelia burgdorferi sensu lato group. Despite this distinction, other resources, including DOID, EFO, and MONDO, align these MeSH entities to a single disease concept using exactMatch or hasDbXref, thereby masking the hierarchical and taxonomic differences preserved in MeSH.

These examples illustrate how equivalent or near-equivalent mapping relations are applied to entities that are modelled with distinct levels of granularity across resources, resulting in structurally divergent representations of clinically related concepts.

4.5. Visualization

A first version of the visualization was developed (shown here as a fixed, simplified snapshot). The interface allows interactive selection of the elements to display, enabling dynamic exploration of the mapping data.

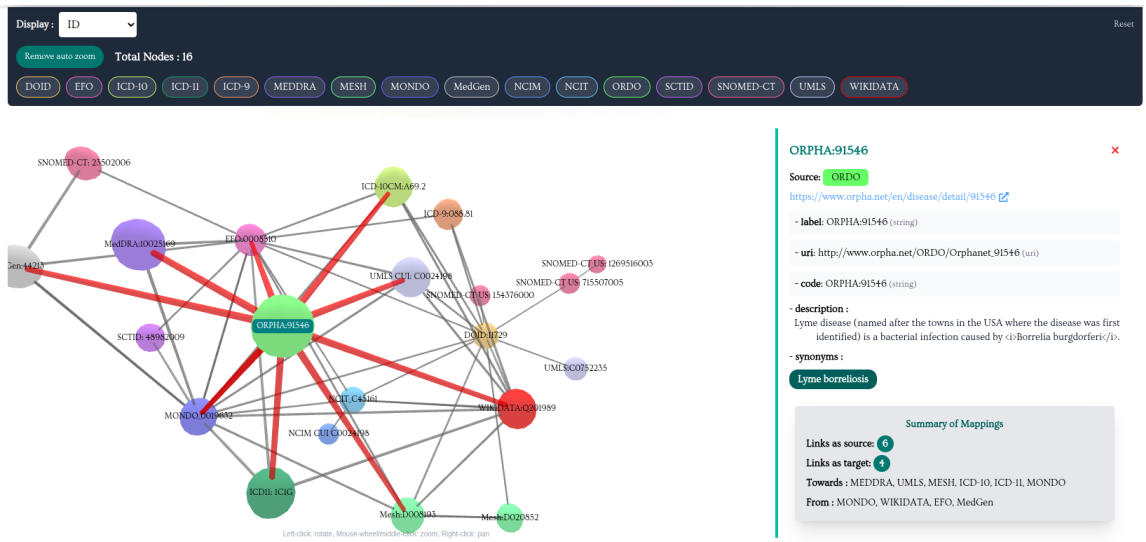

Figure 2: Static view of the visualisation prototype applied to Lyme disease around ORDO

5. Discussion

5.1. What mapping practices reveal beyond ontology coverage

The results show that the majority of biomedical semantic resources rely heavily on oboInOwl:hasDbXref to express cross-resource links [6, 13], while only a subset makes use of SKOS mapping properties to add an explicit semantic layer. This pattern reflects a common distinction in current practices: most resources prioritise identifier-level cross-referencing, whereas fewer explicitly encode the nature of semantic alignment between disease concepts. Although both approaches support interoperability, they imply markedly different levels of semantic commitment and interpretability.

From a theoretical perspective, one might expect more expressive OWL relations such as owl:sameAs or owl:equivalentClass to be used to formalise equivalence or relatedness between disease concepts. In practice, however, such relations are rarely asserted. This observation is consistent with established guidance in ontology engineering and biomedical informatics, which cautions against the routine use of strong equivalence relations outside of well-defined logical identity, due to their strict entailment semantics and the difficulty of justifying exact correspondence between complex biomedical. [30, 31]

As a result, resources tend to favour lighter-weight mapping properties such as oboInOwl:hasDbXref and SKOS relations[19, 20]. These properties offer greater modelling flexibility and are more compatible with heterogeneous disease representations, but they also reduce the explicitness of semantic intent, leaving interpretation largely to downstream users and applications.

Finally, differences in internal modelling practices further complicate cross-resource integration. Resources often define their own internal relations and hierarchical structures to capture disease granularity, which are not always aligned with shared vocabularies or transferable through mappings. Consequently, clinically or biologically meaningful distinctions preserved within individual resources may be partially obscured when integrating disease knowledge across datasets.

5.2. Disease-level consequences of mapping choices: the case of Lyme disease

The disease-focused analysis reveals that mapping choices have tangible consequences when applied to complex and heterogeneous conditions. Diseases are rarely defined as single, perfectly bounded entities, instead, they encompass multiple subtypes, manifestations, or contextual perspectives that coexist within biomedical knowledge. Whether these distinctions remain visible after integration depends largely on how entities are linked across semantic resources.

In this study, we show that Lyme disease is represented across resources through multiple entities reflecting different perspectives, including general disease definitions, clinical subtypes, neurological manifestations, and region-specific classifications. This diversity is consistent with the biological and clinical heterogeneity of the disease [11]. While such modelling choices are legitimate, the mappings connecting these entities determine how this diversity is preserved or obscured in integrated representations.

Several observed cases illustrate how equivalence-oriented mappings may blur clinically or biologically meaningful distinctions. In particular, mappings that align general Lyme disease concepts with specific manifestations or syndromes using skos:exactMatch implicitly assert interchangeability, even when source resources encode hierarchical or subtype relationships. When propagated across resources, these assertions can flatten disease structure and mask internal organisation preserved by more fine-grained models.

These effects are particularly pronounced in the case of Lyme disease where differences in causative agents, clinical presentation, and regional epidemiology are central to interpretation [12]. Variations observed in how resources describe the bacteria responsible for Lyme disease further highlight the importance of semantic precision. From a clinical perspective, treating a patient with Bannwarth syndrome as equivalent to a general Lyme disease concept, or conversely equating Lyme disease with this specific neurological manifestation, may influence how information is interpreted and managed across clinical records and knowledge-based systems.

Overall, these observations emphasise the importance of mapping practices that preserve semantic granularity to support accurate and consistent reuse of disease knowledge.

5.3. Identifier practices and representation as a source of ambiguity

Beyond the choice of mapping properties, the analysis highlights that the representation of external identifiers themselves is a major source of ambiguity and technical challenges [32]. In several resources, external mappings are expressed as string literals rather than as resolvable RDF nodes [17]. When an identifier is encoded as a literal, for example:

```
:someDisease oboInOwl:hasDbXref "MONDO:0019632" .
```

the referenced concept cannot be automatically retrieved or traversed as part of the RDF graph [32]. In such cases, additional processing is required, such as string parsing or the use of BIND operations in SPARQL to reconstruct a resolvable URI. This contrasts with representations where identifiers are expressed as CURIEs or IRIs:

```
:someDisease oboInOwl:hasDbXref MONDO:0019632 .
```

which can be directly resolved through prefix declarations and queried as graph nodes. The latter approach enables automated navigation, federated querying, and reuse across tools without additional transformation steps.

The use of CURIEs is particularly advantageous in ontology editing and exploration tools such as Protégé, where consistent identifier patterns allow automatic resolution and cross-navigation between resources. However, this benefit depends on strict consistency in CURIE formatting. Divergent conventions across resources, such as ICD-10:A69.2 versus ICD10:A69.2, or variations in case sensitivity, break this interoperability and require more normalization. Similar issues arise when multiple string literals encode the same identifier with minor typographical differences, as observed for NCIT:C45161 and NCIt:C45161 within the same resource.

In addition to identifier representation, some resources encode semantic information about mappings through internal annotation practices rather than through standardised mapping properties. For example, qualitative statements such as “exact mapping” or “broad mapping” may be embedded in free-text annotations or resource-specific metadata fields. While such annotations may be informative for human curators familiar with the resource, they are not directly interoperable and cannot be reliably extracted without resource-specific knowledge and manual interpretation. Standard properties such as

skos:exactMatch, skos:broadMatch, and skos:narrowMatch were explicitly designed to convey semantic alignment in a machine-interpretable and interoperable manner [20]. When equivalent information is instead encoded as string-based annotations or resource-specific properties, it undermines the ability to perform systematic extraction and comparison across resources. Users must first identify and interpret these internal conventions before accessing the intended semantic content.

More generally, representing objects as string literals in RDF is discouraged except for purely descriptive annotations intended for human reading [33]. When such literals are used to encode identifiers or semantic relations, they introduce avoidable ambiguity, hinder automated processing, and increase the risk of errors due to inconsistent formatting. Best practices recommend combining human-readable annotations with explicit, resolvable semantic relations whenever possible, ensuring both interpretability and interoperability [32].

5.4. Implications for FAIR data integration and reuse

The findings of this study have broader implications for the reuse and interpretation of biomedical semantic resources, particularly in the context of rare diseases. As these conditions are characterised by limited data availability and highly fragmented knowledge, ontologies and related semantic resources often constitute the primary structured representations available for integration, comparison, and analysis. In such settings, the way diseases are described, connected, and aligned across resources becomes a critical factor in downstream reuse. From a FAIR perspective (Findable, Accessible, Interoperable, Reusable)[4], mappings directly affect the findability and interoperability of disease knowledge by enabling cross-resource discovery, alignment, and reuse. Ambiguous or weakly defined mappings undermine not only interoperability, but also reusability, as their semantic intent cannot be reliably interpreted outside the originating resource.

This issue is further amplified by the growing use of biomedical ontologies in data-driven and AI-based applications, including large language models and hybrid knowledge-driven systems [34, 35]. High-quality AI outputs increasingly depend on structured, well-defined, and interoperable knowledge sources. Ontologies and semantic mappings therefore play a foundational role, not only as background knowledge but as explicit mechanisms for grounding concepts, resolving ambiguity, and connecting heterogeneous data sources. Inconsistent or weakly defined mappings can propagate uncertainty into these systems.

Our results show that, despite the centrality of mappings for interoperability, their semantics and intended use are often implicit, heterogeneous, or resource-specific. While initiatives such as the Simple Standard for Sharing Ontological Mappings (SSSOM) [19] represent a major step forward by promoting structured metadata, provenance, and transparency, this study highlights that metadata alone is not sufficient. Effective interoperability also requires a shared understanding of which mapping properties to use, why they are used, and what semantic commitments they entail.

In practice, the lack of clear, accessible guidance on mapping property selection creates a barrier for both producers and consumers of semantic resources. Curators may default to broadly used but semantically weak properties, while users must invest significant effort to interpret the meaning and reliability of mappings. This challenge is particularly pronounced in complex and semantically unstable disease domains, where subtle distinctions between concepts have important clinical and research implications.

Taken together, these observations suggest the need for clearer conceptual and practical guidance on mapping practices, complementing existing standardisation efforts. Supporting interoperability in rare disease research requires not only structured representations and metadata frameworks, but also explicit, community-oriented guidance that helps stakeholders understand and apply mapping properties in a consistent and meaningful way.

5.5. Role of visualization in supporting mapping interpretation

Several initiatives already support the discovery and reuse of biomedical mappings, yet an important gap remains: it is still difficult to obtain a clear overview of how a disease is represented and mapped across multiple semantic resources, and to understand the impact of differing modelling choices. Existing tools mainly facilitate access to mappings, but provide limited support for analysing their semantic coherence.

For example, OXO aggregates cross-references between identifiers but does not assess semantic correctness or highlight modelling inconsistencies [36]. SEMRA offers programmatic access to SSSOM-formatted mappings and supports retrieval by disease or identifier, but it requires technical expertise and does not explain conceptual divergences between resources [37]. Our approach is therefore complementary: rather than increasing the number of accessible mappings, we focus on analysing how diseases are conceptualised across resources and how this affects interoperability and semantic clarity.

In this context, the prototype visualisation developed in this work addresses a practical gap identified during the analysis. By bringing together disease entities, mapping properties, and resource coverage in a single view, it enables users to explore mapping density and structural discrepancies without requiring expertise in RDF or SPARQL. The visualisation is intended as an exploratory tool to support critical inspection of existing mappings, rather than to resolve inconsistencies or enforce modelling decisions.

This visualisation represents an initial implementation focused on Lyme disease, serving as a foundation for extending the approach to other diseases. It highlights areas where further refinements can improve usability and analytical capabilities. Future developments will aim to support interactive exploration of mappings for any disease and to complement the graph-based view with additional representations, including tables, textual summaries, and a fully developed user interface.

5.6. Limitations and generalizability

his study focuses on explicitly asserted mappings available in downloadable RDF or OWL distributions and accessible metadata services. Mappings that exist only in proprietary systems, restricted-access resources, or undocumented pipelines were not considered. Additionally, no reasoning or inference was applied to derive implicit relations, and the analysis reflects the state of resources at a specific point in time.

While Lyme disease provides a rich and illustrative case, the observed patterns are not expected to be unique to this disease. Similar challenges are likely to arise for other complex conditions characterised by heterogeneous manifestations, evolving definitions, or regional variation. The cascade-based workflow and analytical approach presented here can be applied to other diseases.

While the workflow and the prototype presented here focuses on Lyme disease, their design is intended to be generalizable to other rare diseases, providing a structured approach to explore mappings across multiple semantic resources. One key challenge remains the integration of highly communal resources such as Wikidata, which contains a wealth of biomedical information but does not yet allow straightforward extraction of mappings in a standardised SSSOM format. Automating such extraction could greatly facilitate research with mapping analysis and interoperability in the future.

Additionally, accessing complete, high-quality data across resources remains non-trivial, and initiatives implementing FAIR data points provide complementary strategies to make these data more findable, accessible, interoperable, and reusable. Combining such FAIR-aligned infrastructures with user-friendly visualization tools, as developed in this study, could enhance the exploration, interpretation, and reuse of disease-related mappings for researchers, clinicians, and other stakeholders.

6. Conclusion

This study demonstrates that, even for a well-established disease such as Lyme disease, mappings across major biomedical semantic resources remain limited in semantic expressivity. Most resources rely on a small set of properties (primarily oboInOwl:hasDbXref, skos:closeMatch, and skos:exactMatch,) which

often provide insufficient information to assess how closely two disease concepts are intended to align. As a result, conceptual differences between resources may remain implicit or be inadvertently flattened during integration.

These limitations are particularly consequential in the rare disease context, where knowledge is inherently fragmented and interoperability is essential for FAIR data reuse. Our analysis shows that identifying and interpreting disease mappings remains challenging, especially for non-technical users, requiring manual exploration across multiple resources and resolution of inconsistencies that are rarely made explicit.

The interactive visual prototype presented in this work illustrates how mapping choices and modelling differences can become more intelligible when explored through a graph-based representation. While currently limited to Lyme disease and a predefined set of resources, the underlying workflow is generalisable and can be extended to other diseases. Integrating structured mapping metadata, such as those promoted by initiatives like SSSOM, offers a promising direction for improving transparency and interpretability.

Overall, our results highlight the need for clearer guidance and more consistent use of mapping relations, complemented by accessible exploration tools, to support reliable integration of disease knowledge across biomedical semantic resources. Addressing these issues is a necessary step toward making semantic interoperability practically usable for research and clinical applications, particularly in the rare disease domain.

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Declaration on Generative AI

During the preparation of this work, the authors used Chat-GPT-4 and DeepL in order to: Grammar and spelling check. After using these tools/services, the authors reviewed and edited the content as needed and takes full responsibility for the publication's content.

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A. Online Resources

The source code, datasets, and supplementary materials used in this study are available on github [39].